

Slopes of lines



- 1 The ratio of the vertical change to the horizontal change for any two points on the line.
- 2 $\frac{\text{Vertical rise}}{\text{Horizontal run}}$
- 3 Run
- 4 The line is horizontal.
The change in y is equal to zero.
- 5 The line is vertical.
- 6 No
- 7 a 3 b -3
- 8 a $-\frac{3}{2}$ b -1
- 9 0
- 10 a $-\frac{9}{5}$ b $-\frac{9}{5}$ c Yes
- 11 $t = 15$
- 12 $c = -\frac{12}{7}$
- 13 -1
- 14 $\frac{1}{2}$
- 15 $d = 16$

16

a undefined

b undefined



c defined

d defined

e undefined

f defined

g undefined

h defined

17

a 0

b 0

c undefined

18

a C

b B

19

a 0

b undefined

20

a 2

b 4

c $\frac{1}{2}$

21

a -4

b 1

c -4

22

a -4

b Decreasing

23

a Positive

b Negative

24

a $\frac{2}{3}$ b $\frac{1}{5}$

25

a $-\frac{5}{2}$ b $\frac{1}{10}$

26

a -1

b 1

c -1

27

a -0.76

b -0.35

c Ski run A

28 13.2 cm



Additional questions

29

a Slope = 1

b Slope = 2

c Slope = $\frac{1}{2}$

d Slope = $-\frac{1}{4}$

e Slope = -1

f Slope = $\frac{3}{5}$

30

a Slope = 9

b Slope = 3

c Slope = 7.5

d Slope = $-\frac{7}{5}$